

Detailed Model Analysis for Betel Leaf Disease Classification

CSE366 – Sec 01 – Fall
2025



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Introduction & Problem Statement

- **The Crop:** Betel Leaf (*Piper betle* L.) is vital cash crop in South Asia.
- **The Problem:** High susceptibility to diseases (Bacterial Spot, Foot Rot, Fungal Spot).
- **Current Solution:** Manual visual inspection.
- **Limitations:** Subjective, labor-intensive, prone to human error, requires experts.
- **Proposed Solution:** Automated, precision driven diagnostic tool using Deep Learning.

Research Objectives

- **Benchmark:** Evaluate standard Transfer Learning models against Custom CNNs.
- **Innovate:** Develop and test a Hybrid CNN-Swin Transformer architecture.
- **Analyze:** Compare the impact of Attention Mechanisms (CBAM, SE-Blocks).
- **Validate:** Ensure robustness on imbalanced datasets and visualize decisions using XAI (Grad-CAM).

Evaluation Metrics & Protocol

- **Dataset:** 1,000 Images (Balanced) + Secondary Imbalanced Test Set.
- **Hardware:** CUDA-enabled GPPU (Tesla P100/T4)
- **Key Metrics Recorded:**
 - Test Accuracy (%)
 - Training Time
 - Loss Convergence
 - Confusion Matrix Performance
- **Preprocessing:** Resize (224x224), Normalization (ImageNet stats), Augmentation (Rotation, Flip, Color Jitter).

Dataset Overview

Healthy Leaf



Dried Leaf



Fungal Brown Spot Disease



Bacterial Leaf Disease



- **Source:** Betal Leaf Image Dataset from Bangladesh.
- **Class (4):**
 1. Bacterial Leaf Disease
 2. Dried Leaf
 3. Fungal Brown Spot Disease
 4. Healthy Leaf
- **Data Splits:**
 - Dataset 1 (Balanced): 1,000 images (250/class). Used for training.
 - Dataset 2 (Imbalanced):** Real-world distribution (High variance in class counts). Used for robustness testing.
- **Preprocessing:** Resize (224x224), Normalization (ImageNet stats), Augmentation (Rotation, Flip, Color Jitter)

Methodology – The Models

Category 1: Transfer Learning Baselines

InceptionV3, ResNet50, MobileNetV2,
DenseNet121,
Xception, VGG16, AlexNet, EfficientNet-B0

Category 2: Custom Architectures

Custom Sequential CNN (Lightweight convolutions).
BLCNN (Depth-wise separable convolutions).

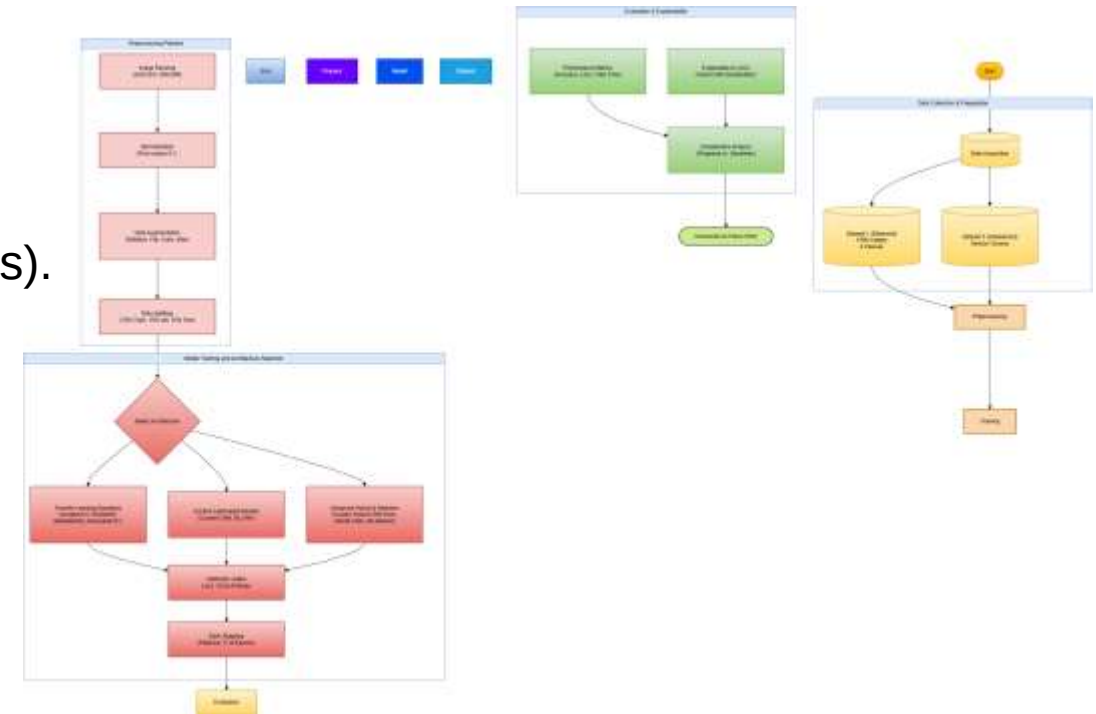
Category 3: Advanced Hybrid & Attention

Hybrid CNN-Swin Transformer: Combines local features

(CNN) + global context (Transformer).

CBAM-CNN: Spatial + Channel Attention.

SE-ResNet: Squeeze-and-Excitation blocks.



Proposed Architecture: Hybrid CNN-Swin Transformer

Key Components:

CNN Stem:

Extracts low-level textural features (edges, shapes).

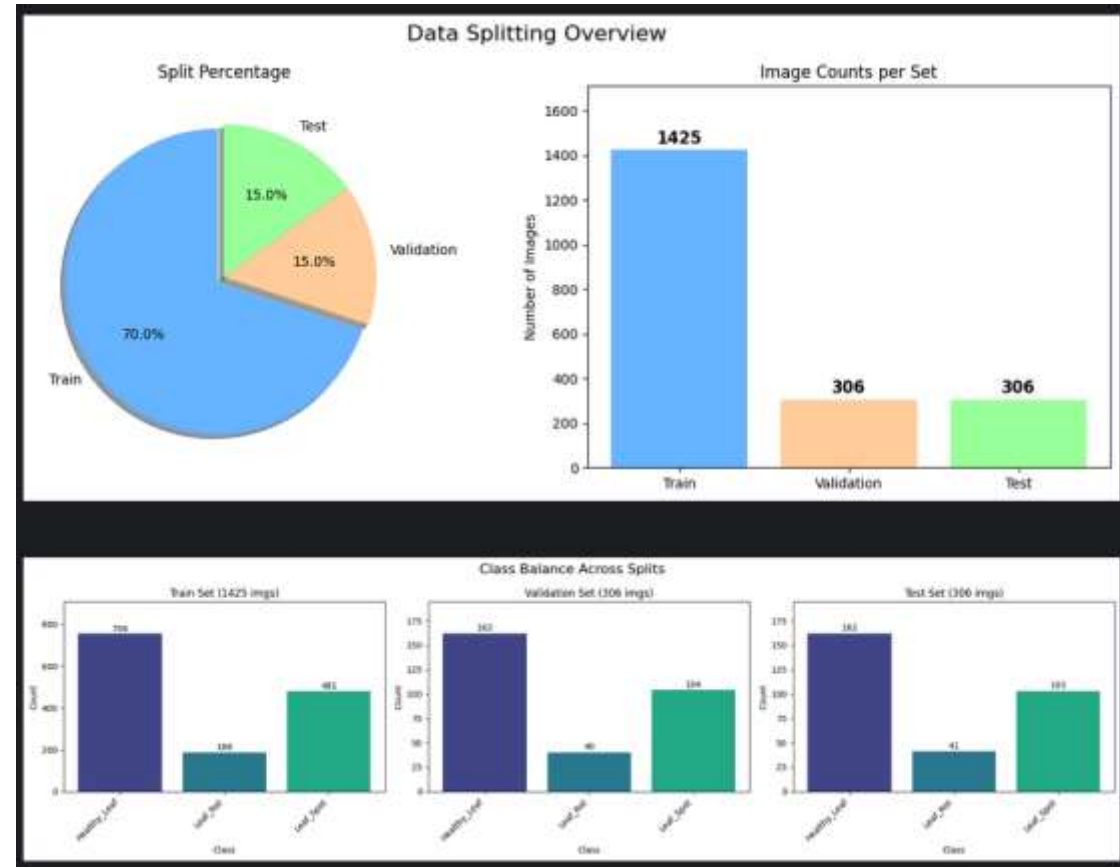
Swin

Transformer Blocks:

Captures long-range dependencies and global context using shifted windows.

Classification

Head: Global Average Pooling +Dense Layers.



Picture of data splitting on the Dataset
2.

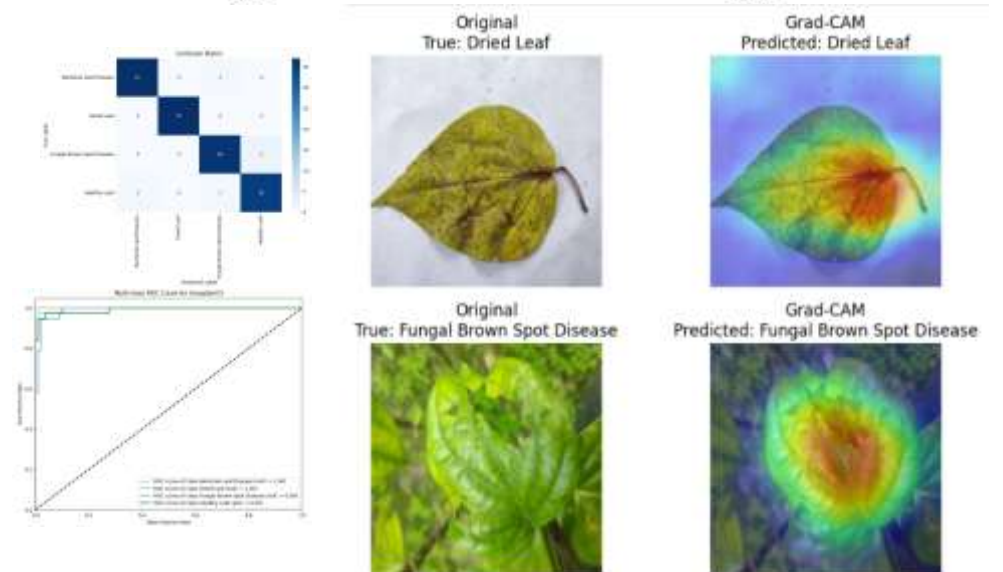
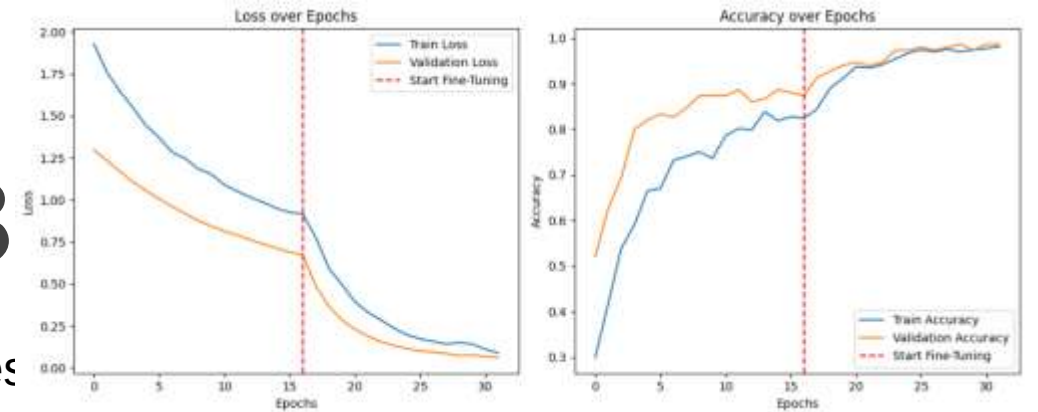
Experimental Result

Section 1: Transfer Learning Baselines

Evaluating industry-standard pre-trained architectures.

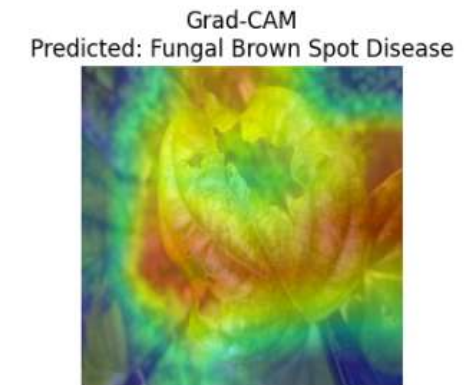
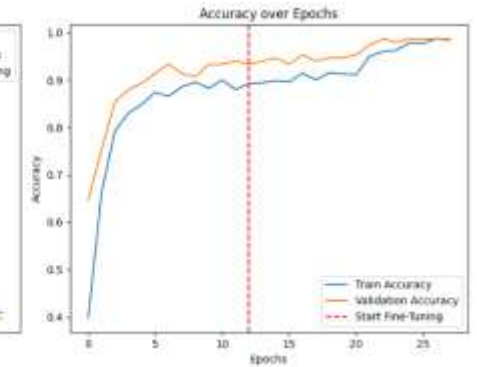
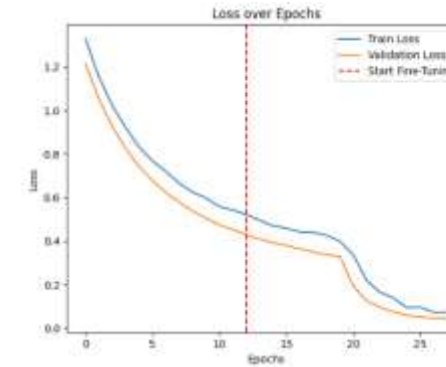
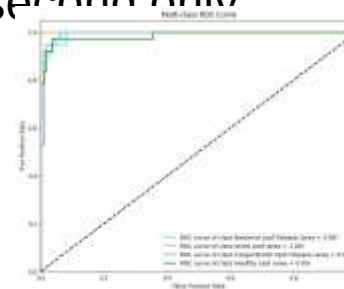
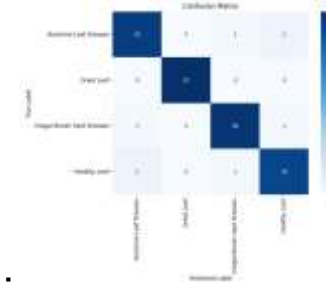
Model 1 – Inception V3

- **Architecture Type:** Deep CNN with Inception Modules (Multi-scale processing).
- **Training Configuration:** Transfer Learning (ImageNet weights), Fine-tuned.
- **Performance:**
 - **Accuracy: 96.7%** (Top Performer in Baselines)
 - **Training Time:** ~3.09 mins
- **Analysis:**
 - Excellent generalization.
 - The multi-scale filters allowed it to detect disease spots of varying sizes effectively.
 - **Verdict:** Best standard CNN for this task.



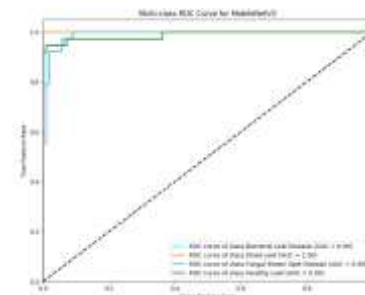
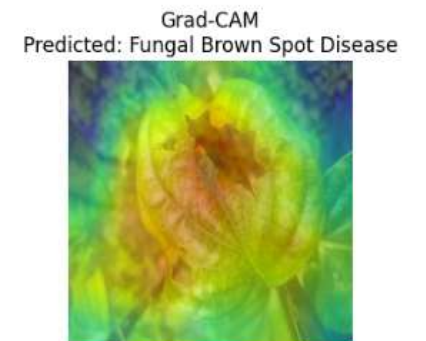
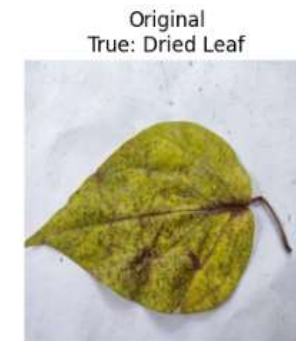
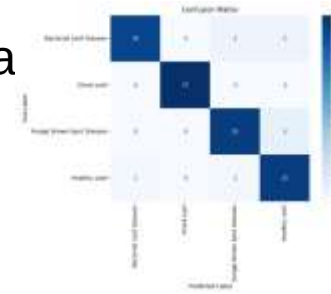
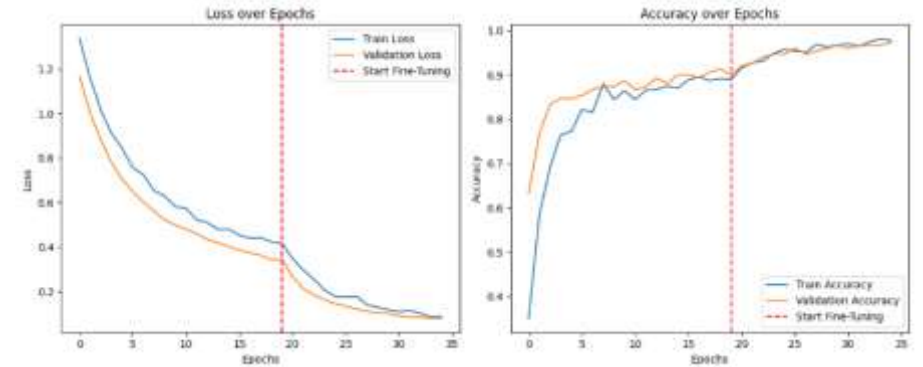
Model 2 – ResNet50

- **Architecture Type:** Residual Network (50 layers) with skip connections.
- **Training Configuration:** Pre-trained, Early Stopping enabled.
- **Performance:**
 - **Accuracy:** 95.0%
 - **Training Time:** ~1.56 mins
- **Analysis:**
 - Very fast convergence.
 - Solved vanishing gradient issues effectively.
 - **Verdict:** Highly reliable and efficient, second only to InceptionV3 among baselines.



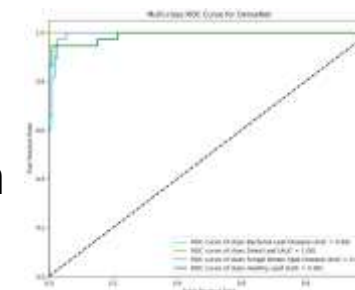
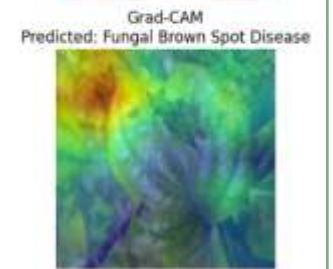
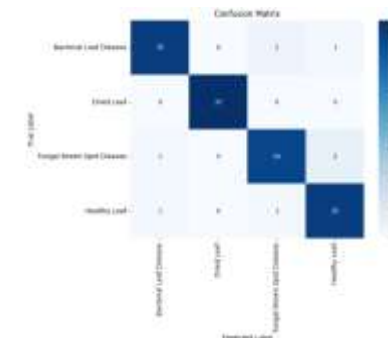
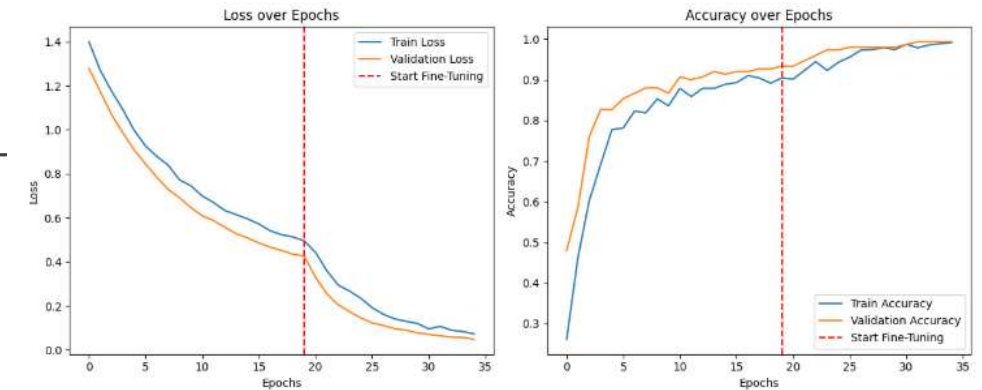
Model 3 – MobileNetV2

- **Architecture Type:** Lightweight CNN with Depth-wise Separable Convolutions.
- **Training Configuration:** Standard Transfer Learning
- **Performance:**
 - **Accuracy:** 94.0%
 - **Training Time:** ~1.44 mins
- **Analysis:**
 - Designed for mobile devices; extremely low latency.
 - Performed comparably to the much heavier RedNet50
 - **Verdict:** Ideal candidate for mobile app deployment if resources are tight.



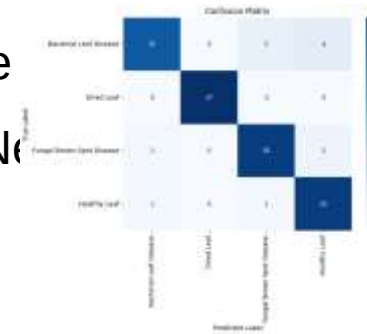
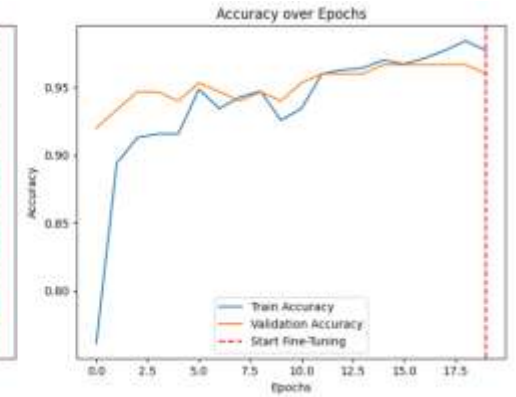
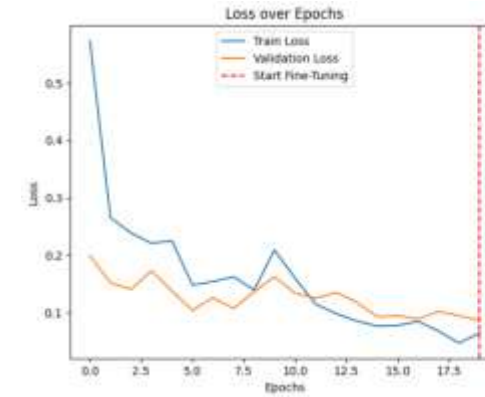
Model 4 – DenseNet121

- **Architecture Type:** Densely Connected Convolutional Network.
- **Training Configuration:** Fine-tuned (Unfrozen deeper layers).
- **Performance:**
 - **Accuracy:** 94.0%
 - **Training Time:** ~131 seconds (Higher computational cost).
- **Analysis:**
 - Feature reuse makes it powerful but computationally expensive.
 - Did not significantly outperform MobileNet despite higher complexity.
 - **Verdict:** Good accuracy, but diminishing returns on efficiency.



Model 5 – AlexNet

- **Architecture Type:** Classic shallow CNN (Large filters).
- **Training Configuration:** Pre-trained on ImageNet.
- **Performance:**
 - **Accuracy:** 92.0%
 - **Training Time:** ~50 seconds (Very Fast).
- **Analysis:**
 - **Surprise Finding:** Outperformed modern architectures like EfficientNet and Xception.
 - Simpler inductive biases worked well for the specific morphological features of betel leaves.
 - **Verdict:** Proof that older, simpler models can still be effective on smaller datasets.

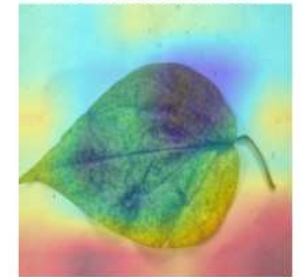


--- Grad-CAM Visualization ---

Original
True: Dried Leaf



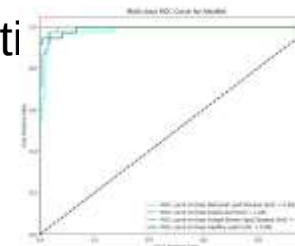
Grad-CAM
Predicted: Dried Leaf



Original
True: Fungal Brown Spot Disease

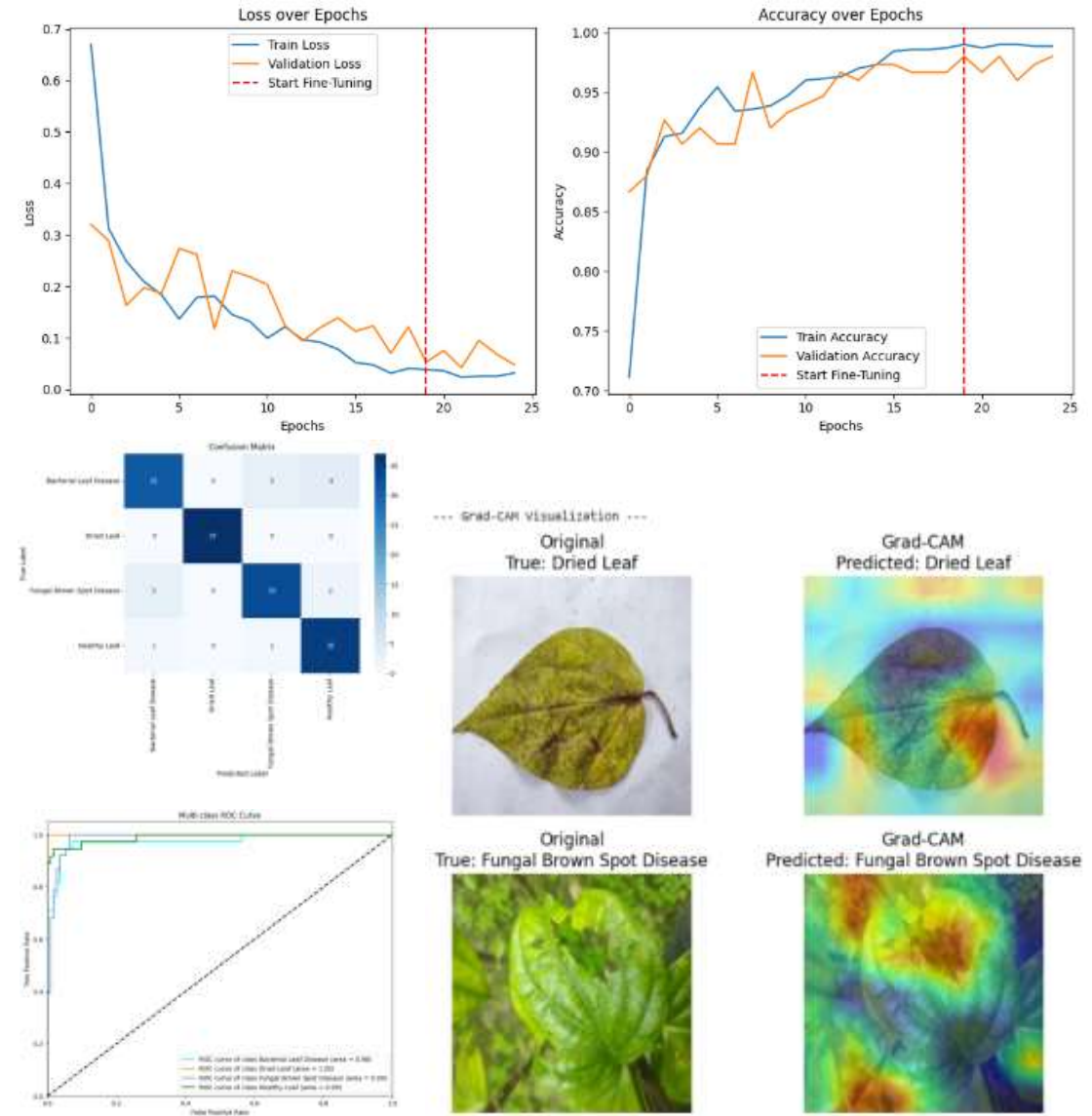


Grad-CAM
Predicted: Fungal Brown Spot Disease



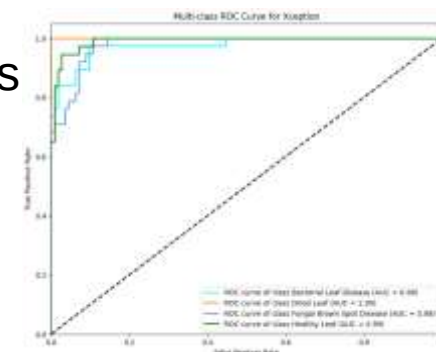
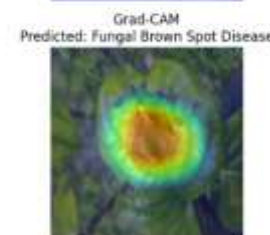
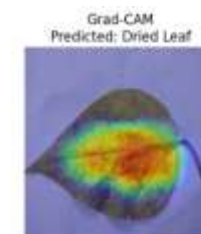
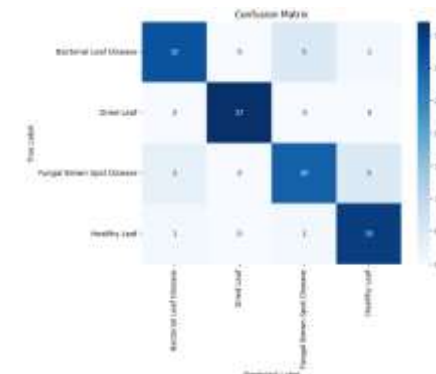
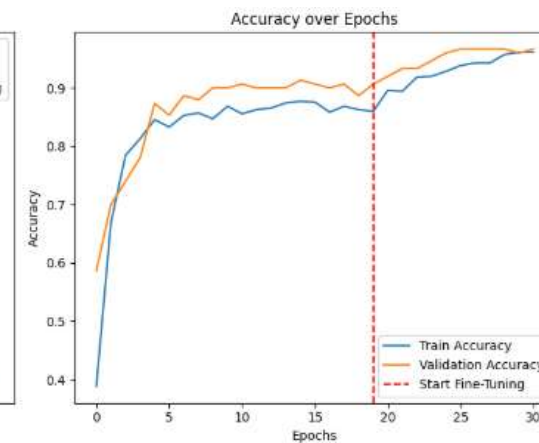
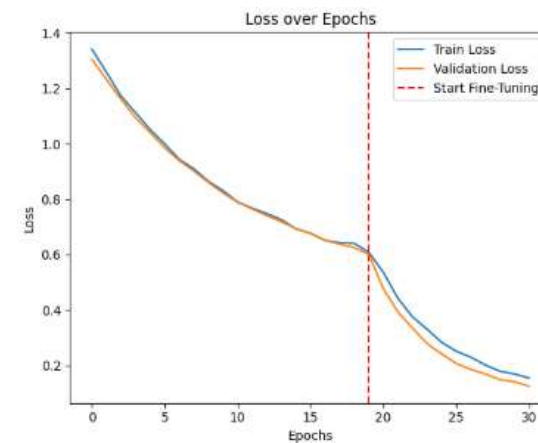
Model 6 – VGG16

- **Architecture Type:** Very Deep Convolutional Network (16 layers).
- **Training Configuration:** Transfer Learning.
- **Performance:**
 - **Accuracy:** 91.0%
 - **Training Time:** ~2.19 mins
- **Analysis:**
 - High parameter count made it slower to train than ResNet.
 - Solid feature extractor, but prone to overfitting without heavy regularization.
 - **Verdict:** Decent performance, but outdated efficiency.



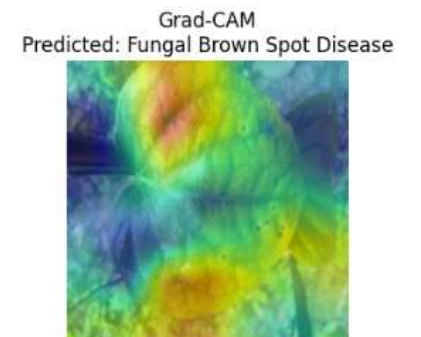
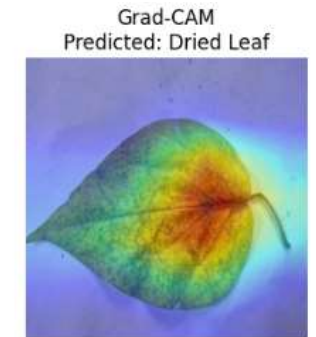
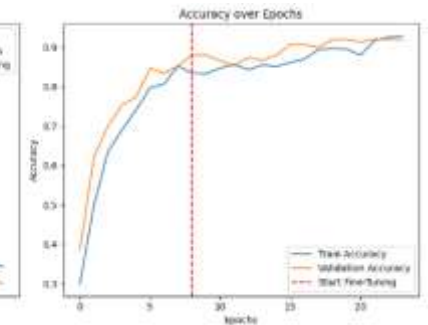
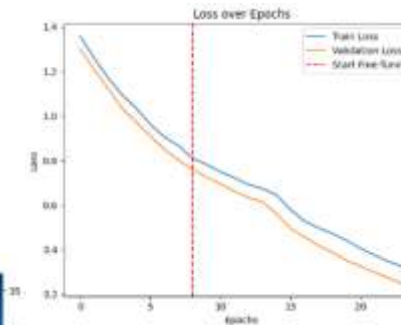
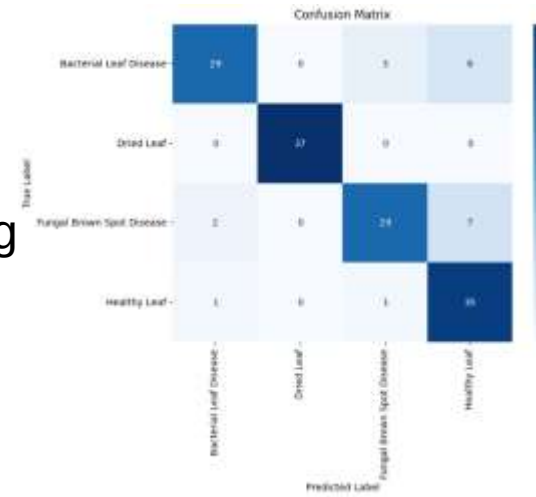
Model 7 – Xception

- **Architecture Type:** Extreme Inception (Depth-wise separable convs).
- **Training Configuration:** Fine-tuned with Early Stopping.
- **Performance:**
 - **Accuracy:** 89.0%
 - **Training Time:** ~242 seconds (Slowest among baselines).
- **Analysis:**
 - Struggled to generalize compared to InceptionV3.
 - High training time suggested optimization difficulties on this specific dataset.
 - **Verdict:** Underperformed relative to expectations.



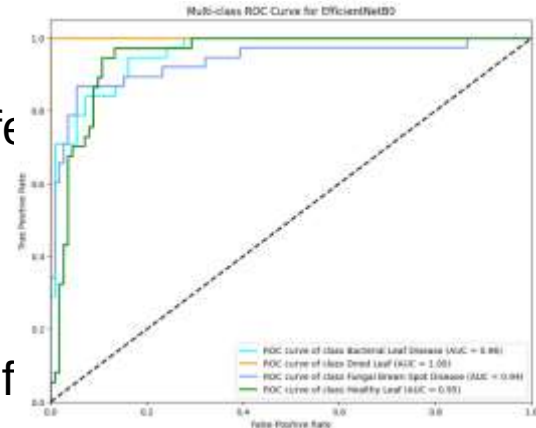
Model 8 – EfficientNet-B0

- **Architecture Type:** Compound Scaling CNN.
- **Training Configuration:** Transfer Learning.
- **Performance:**
 - **Accuracy:** 87.0%
 - **Training Time:** ~1.16 mins
- **Analysis:**
 - Lowest performance among transfer learning models.
 - Likely required more extensive hyperparameter tuning or a larger dataset to leverage its scaling capabilities.
 - **Verdict:** Not suitable for this specific dataset configuration.



Original
True: Fungal Brown Spot Disease

Grad-CAM
Predicted: Fungal Brown Spot Disease

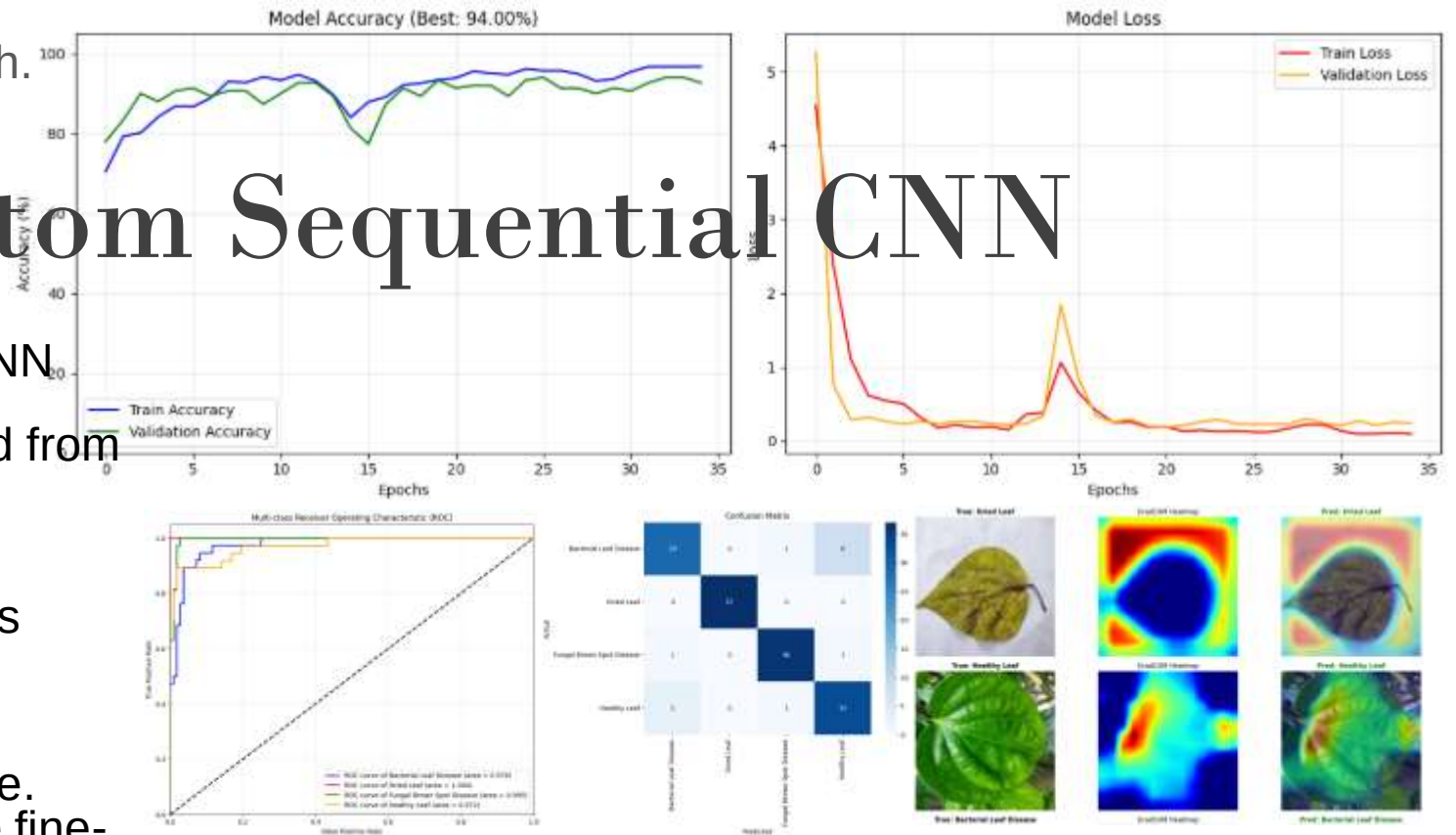


Section 2: Custom Architectures

Evaluating models built from scratch.

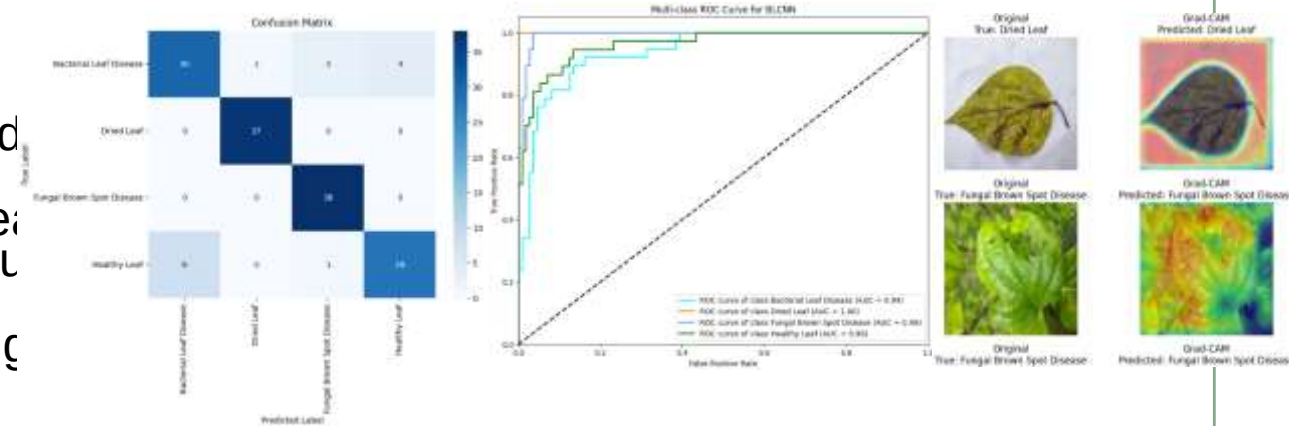
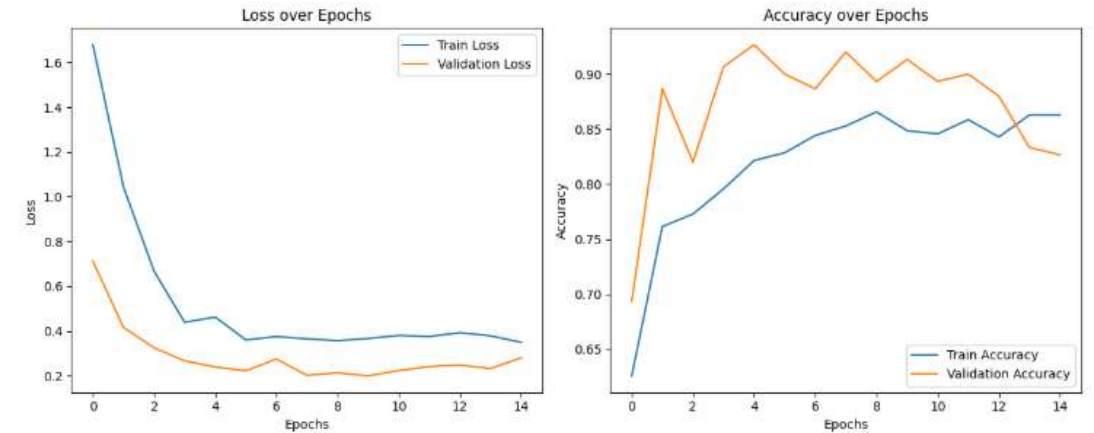
Model 9 – Custom Sequential CNN

- **Architecture Type:** Standard CNN (Conv2D -> ReLU -> MaxPool).
- **Training Configuration:** Trained from scratch (35 Epochs).
- **Performance:**
 - **Accuracy:** 90.0%
 - **Training Time:** ~20 seconds (Fastest).
- **Analysis:**
 - Strong baseline performance.
 - Lacked the ability to capture fine-grained textures distinguishing bacterial vs. fungal spots.
 - **Verdict:** Good starting point, but limited.



Model 10 – BLCNN (Betel Leaf CNN)

- **Architecture Type:** Domain-specific custom architecture (based on literature).
- **Training Configuration:** Trained from scratch.
- **Performance:**
 - **Accuracy:** 88.87%
 - **Training Time:** ~31 seconds.
- **Analysis:**
 - Performed slightly worse than the standard Custom CNN.
 - Indicates that the specific architectural tweaks from literature didn't transfer perfectly to our dataset split.
 - **Verdict:** Less effective than standard design

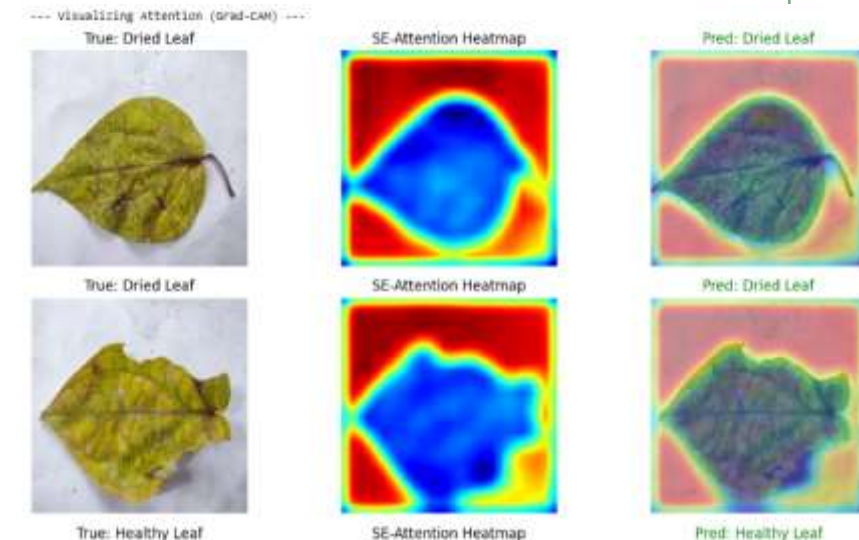
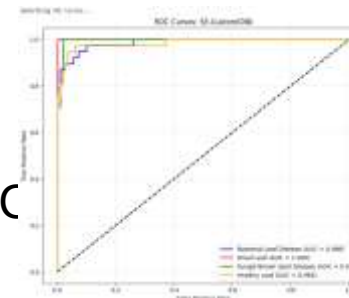
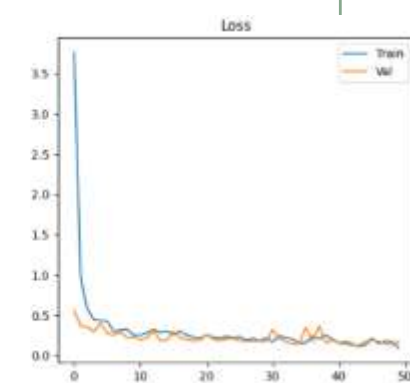
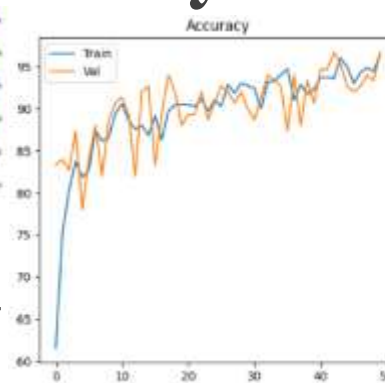
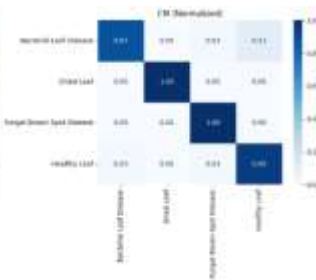
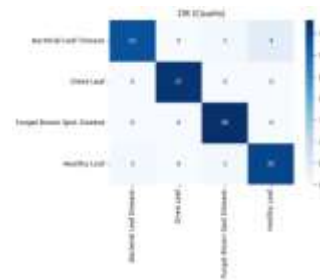


Section 3: Advanced Attention & Hybrid Models

The State-of-Art Contenders.

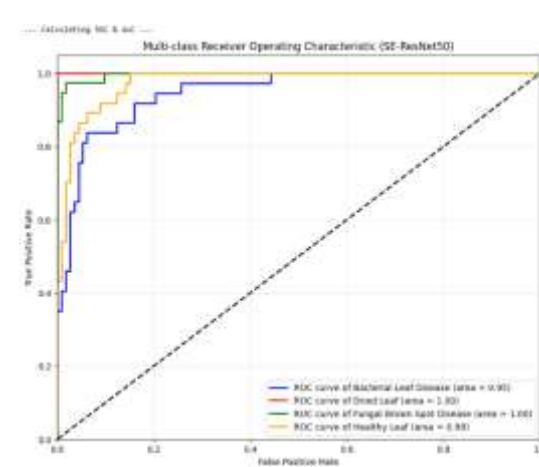
Model 11 – SE-Custom CNN

- **Architecture Type:** Custom CNN + Squeeze-and-Excitation (SE) Blocks.
- **Mechanism:** Channel-wise attention (Recalibrates feature maps).
- **Performance:**
 - **Accuracy: 95.0%**
 - **Training Time:** ~6 minutes 47 seconds
 - **Improvement:** +5% over standard Custom C
- **Analysis:**
 - Demonstrated that channel attention significantly boosts feature extraction in lightweight models.
 - Lacked the ability to capture fine-grained textures distinguishing bacterial vs. fungal spots.
 - **Verdict:** Excellent balance of custom design and high accuracy.



Model 12 – SE-ResNet

- **Architecture Type:** ResNet backbone + SE Blocks.
- **Performance:**
 - **Accuracy:** 88.0%
 - **Training Time:** ~7 minutes 26 seconds
- **Analysis:**
 - Surprisingly underperformed compared to the SE-Custom CNN.
 - Suggests that simply adding attention to a heavy backbone doesn't guarantee success without careful tuning.
 - **Verdict:** Disappointing implementation compared to custom SE.

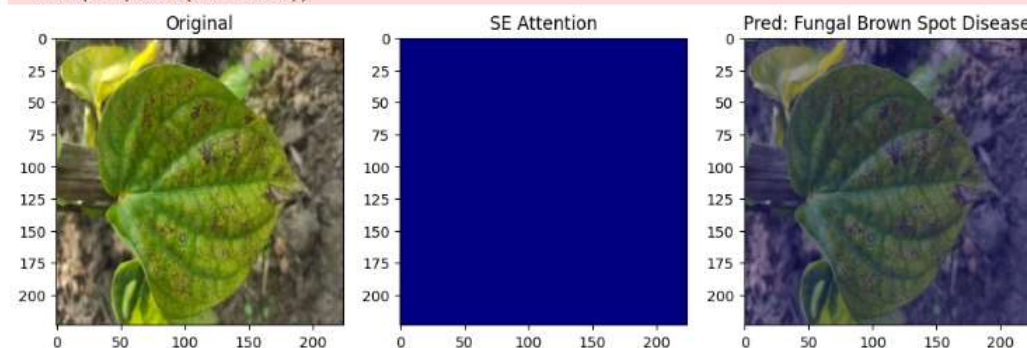


SE-ResNet50 Confusion Matrix

	Bacterial Leaf Disease	Dried Leaf	Fungal Brown Spot Disease	Healthy Leaf
Bacterial Leaf Disease	31	2	1	3
Dried Leaf	0	38	0	0
Fungal Brown Spot Disease	2	2	34	0
Healthy Leaf	8	0	0	29

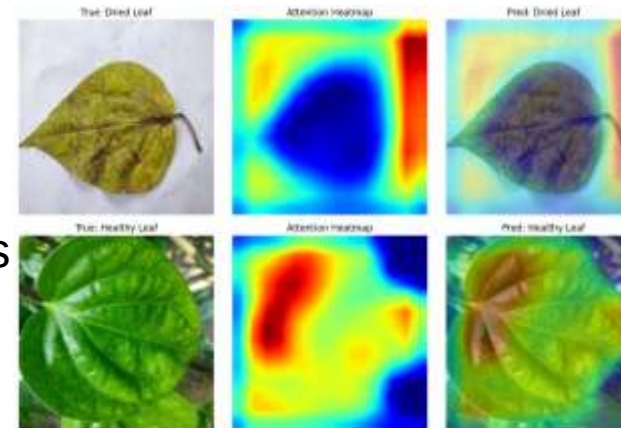
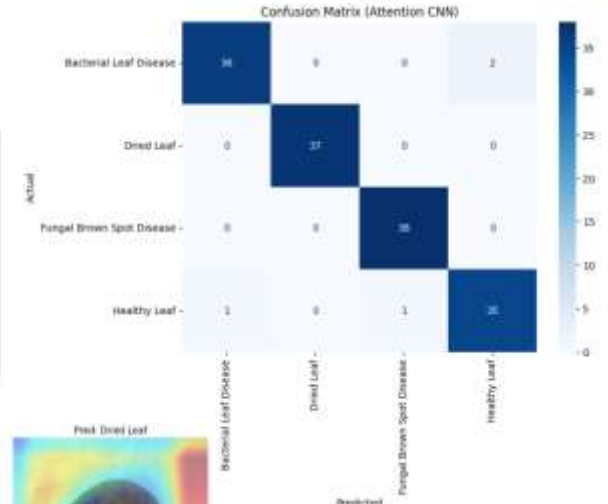
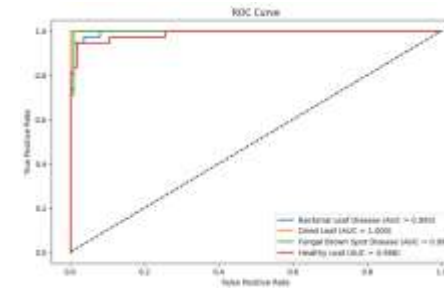
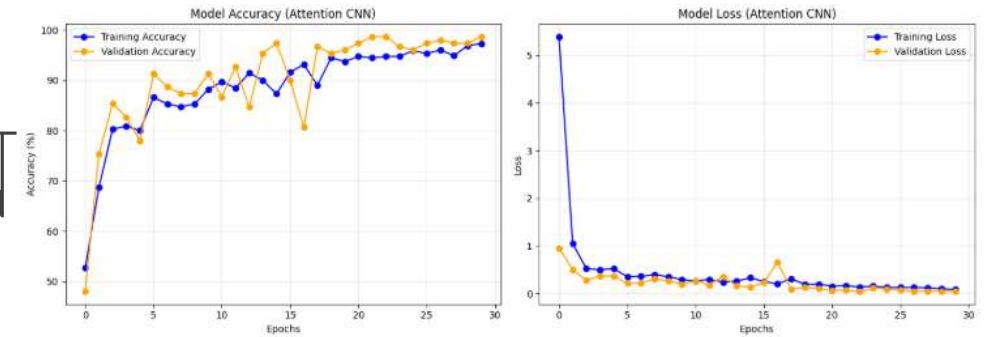
--- Grad-CAM Explainability ---

/tmp/ipykernel_20/1968706771.py:555: RuntimeWarning: invalid value encountered in cast
heatmap = np.uint8(255 * heatmap)



Model 13 – CBAM-CNN

- **Architecture Type:** CNN + Convolutional Block Attention Module.
- **Mechanism:** Spatial Attention (“Where to look”) + Channel Attention (“What to look for”).
- **Performance:**
 - **Accuracy:** 97.33%
 - **Training Time:** ~25 seconds.
- **Analysis:**
 - Outperformed all pure CNNs (including InceptionV3).
 - Spatial attention helped the model ignore background noise and focus purely on leaf lesions
 - **Verdict:** The Best pure CNN-based attention model.



Model 14 – Hybrid CNN-Swin Transformer

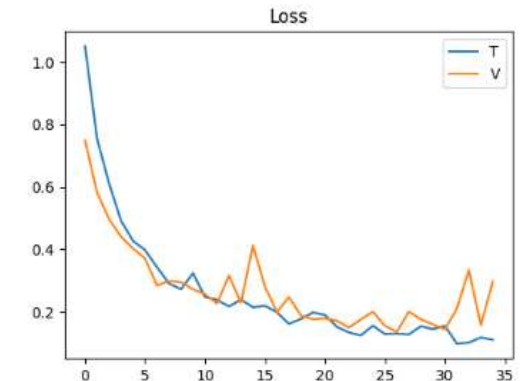
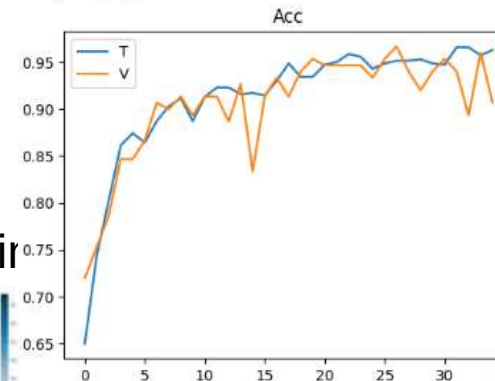
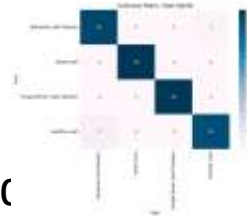
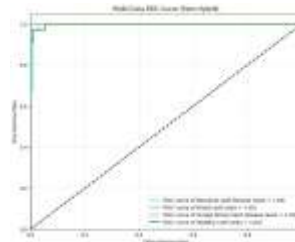
- **Architecture Type:** CNN Stem (Local features) + Swin Transformer Blocks (Global context).

- **Performance:**

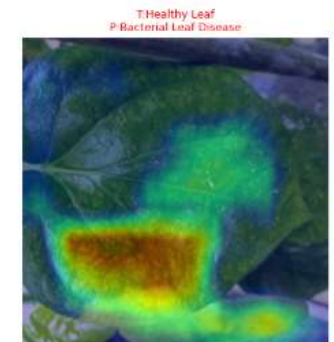
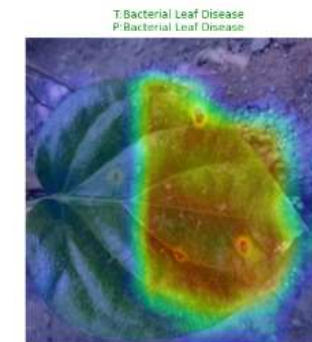
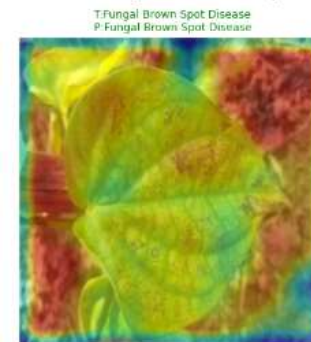
- **Accuracy (Dataset 1):** 98.0%
- **Training Time:** ~ 8 minutes 31 seconds
- **Accuracy (Dataset 2 - Imbalanced):** 93.0%

- **Analysis:**

- **Synergy:** CNN captures edge; Transformer captures the holistic leaf structure.
- **Robustness:** Maintained high accuracy even on the difficult, imbalanced dataset.
- **Verdict: State-of-the-Art Performance.** The recommended model for deployment.



--- Visualizing Transformer Output Attention ---



Robustness & Generalization

Tested on **Dataset 2 (Imbalanced)**.

Challenge : Standard models Transformer maintained 93% accuracy on the imbalanced dataset.

Result: The Hybrid Swin Transformer maintained 93% accuracy on the imbalanced dataset.

Conclusion: The Hybrid model is not just accurate; it is robust for real-world scenarios.





Discussion – Architecture Insights

InceptionV3 : Best pure CNN. Multi-scale processing helped detect disease spots of varying sizes.

AlexNet (92%) : Surprisingly beat modern EfficientNet (87%).

Reason : Simpler inductive biases (larger filters)

worked better for this specific small dataset than

complex compound scaling.

Attention Mechanisms : CBAM and SE blocks consistently improved performance over base²⁴ model by focusing on “where” the disease is.

Explainable AI (Grad-CAM)

- Visuals.
- Goal: Open the “Black Box.”
- Observation:
- Hybrid/CBAM: Heatmaps focus tightly on lesions and necrotic spots.
- Weaker Models: Validates clinical reliability for precision agriculture.

Conclusion & Future Work

- **Conclusion :**

InceptionV3 is the most efficient standard model.

Hybrid CNN-Swin Transformer is the state-of-the-art (98%) for this task.

Integrating local and global attention is key for plant disease diagnosis.

- **Future Work :**

Mobile Deployment : Model quantization for edge devices.

Segmentation : Moving from classification to segmentation (U-Net) to measure infection severity.

Dataset Expansion : More lighting conditions and disease types.

Final Comparative Summary

Rank	Model Name	Test Accuracy	Architecture Type	Training Time
1	Hybrid CNN-Swin Transformer	98.00%	Hybrid Transformer	8m 31s
2	CBAM-CNN	97.33%	Attention CNN	~25s
3	InceptionV3	97.00%	Transfer Learning	3.09m
4	SE-Custom CNN	95.00%	Attention CNN	6m 47s
5	ResNet50	94.00%	Transfer Learning	1.56m
6	MobileNetV2	94.00%	Transfer Learning	1.44m
7	DenseNet121	94.00%	Transfer Learning	2.19m
8	AlexNet	91.00%	Transfer Learning	~50s

Thank you

Reference of the dataset

- **Dataset 1 (Balanced)** link:
<https://data.mendeley.com/datasets/g7fpgj57wc/2>
- **Dataset 2 (Imbalanced)** link:
<https://data.mendeley.com/datasets/vpzknzjty/1>